MILESTONE 1: DATA ANALYSIS

Project: Portfolio Risk-Return Optimization
Roll No: IITMD_BA_2501070_FA-01
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1. **Dataset Overview:**
   The analysis used Nifty 50 stock market data from Kaggle, covering the period 2016-2020 (5 years). After data cleaning, 50 stocks with complete trading data were analyzed. The dataset contained daily OHLC prices, volume, and trading information.

Data Cleaning Steps:

- Filtered data to 2016-2020 period
- Forward filled missing values
- Removed duplicate records
- Calculated daily returns for all stocks

The dataset comprised 60,000 daily trading records after filtering to 2016-2020.

2. **Key Findings:**

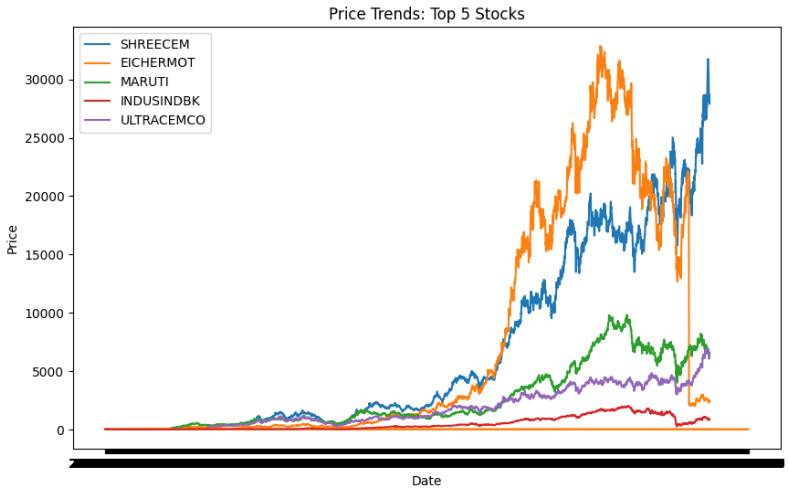
Analysis of 50 Nifty stocks revealed significant performance variation:

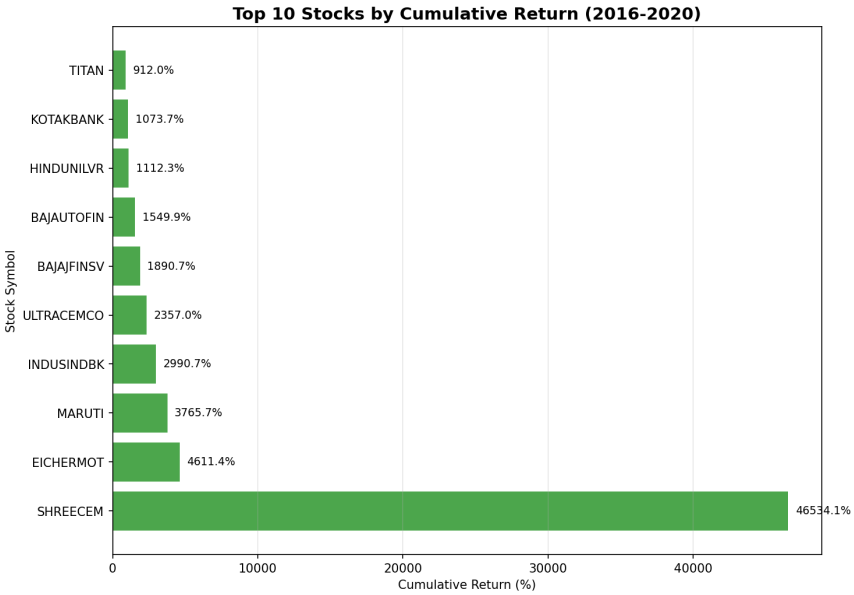
- Best Performer: SHREECEM with +46,534% cumulative return
- Worst Performer: WIPRO with -83.25% cumulative return
- Average Return: 4,428% across all selected stocks
- Mean Daily Return: 0.089%

The distribution of daily returns showed normal distribution with slight positive skew, indicating overall market growth during this period.

3. **Top Performers:**

Rank	Stock	Sector	Cumulative Return	Annualized Return	Volatility
1	SHREECEM	Cement	46,534%	46.5%	38.9%
2	EICHERMOT	Auto	4,611%	41.5%	48.8%
3	MARUTI	Auto	3,766%	30.8%	34.4%



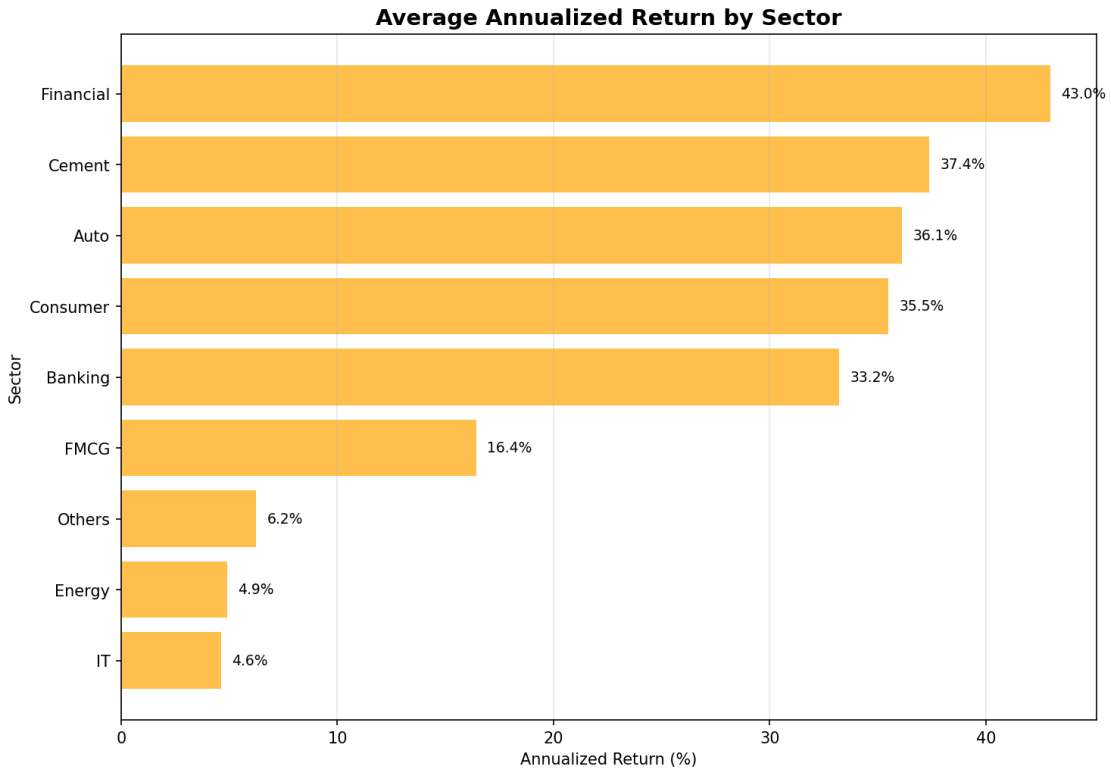


4. Sector-Wise Performance:

Analysis across 9 sectors showed Financial and Auto sectors leading:

- Financial: 43.4% avg annualized return
- Cement: 37.4% avg annualized return
- Auto: 36.1% avg annualized return

IT and Energy sectors showed lower but stable returns during this period.



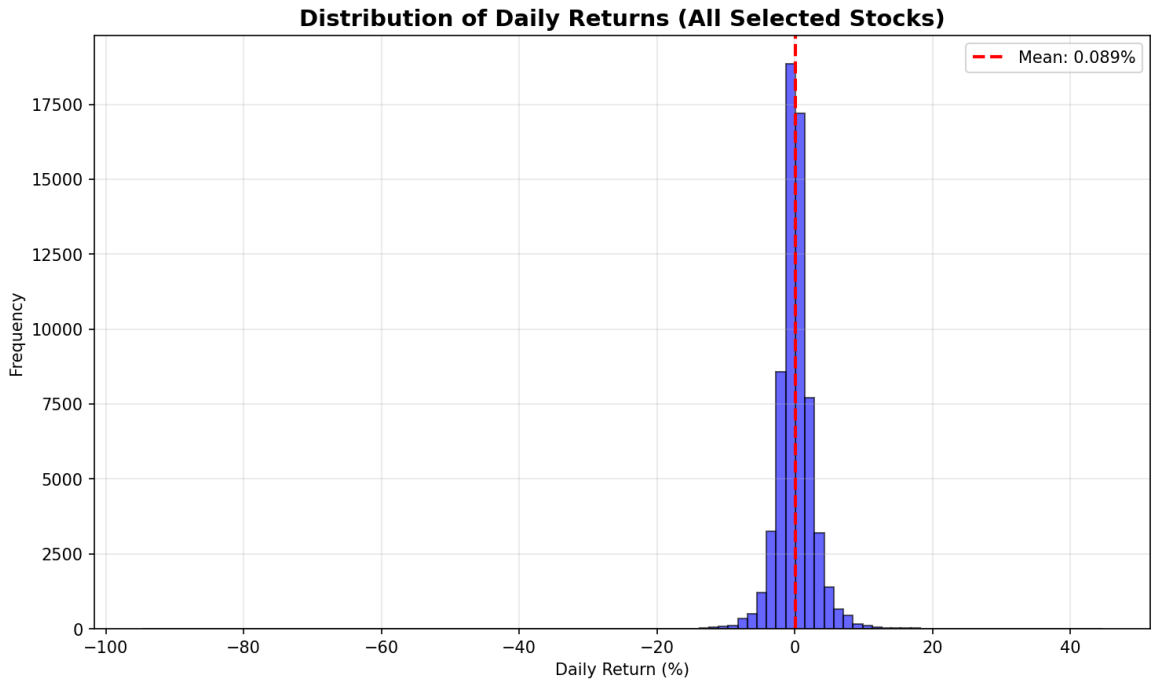
5. Stock Selection Rationale:

15 stocks were selected ensuring:

- Diversification: 9 sectors represented
- Performance Mix: Top 10 performers + Bottom 5 performers
- Sector Balance: Minimum 4 major sectors covered

Selected Portfolio Stocks: SHREECEM, EICHERMOT, MARUTI, INDUSINDBK, ULTRACEMCO, BAJAJFINSV, BAJAUTOFIN, HINDUNILVR, KOTAKBANK, TITAN, IOC, UNIPHOS, ITC, INFOSYSTCH, WIPRO

This selection provides exposure to high-growth (cement, auto) and defensive (FMCG, IT) sectors.



Conclusion:

The exploratory data analysis identified 15 stocks with diverse risk-return profiles suitable for constructing Conservative, Balanced, and Aggressive portfolios in Milestone 2. The 2016-2020 period showed strong equity market performance with sector rotation opportunities.